SemanticSplat / Beyond Proximity

Progress Report and Next Steps

Status date: 2026-07-06 | Branch: codex/research-upgrade-baselines-datasets | Frozen evidence commit: 8700f26

Short answer: the project has a credible workshop-ready semantic-search prototype with reproducible graph-vs-flat evidence. The next scientific step is to close the quality gap and add public-dataset/fair-baseline evidence before aiming at a stronger main-conference submission.

Recommended target	Why
TwinWorld @ ECCV 2026	Best topical fit for digital twins, semantic 3D, 3DGS, scene understanding, and honest workshop-scale evidence.
WACV 2027 Round 2	Best next main-conference path after adding embedding retrieval, calibrated pruning, public-data evaluation, and stronger failure analysis.
Avoid AAAI 2027 main for now	Current evidence is too small and too systems-oriented; graph hit@k is lower than flat under the current lexical ranker.

1. What We Completed

- Integrated the Week 4 research branch into a reproducible evidence state and continued on codex/research-upgrade-baselines-datasets.
- Updated the project truth state: five captured pilot scenes now contain 97 captured views and 1,066 semantic items.
- Regenerated the primary five-scene graph-vs-flat benchmark on 150 queries with 125 verified-view-label queries.
- Added graph-pruning controls and tests for tighter, default, broad, and affordance-aware traversal variants.
- Added ablation and scaling stress scripts plus checked-in evidence outputs.
- Reran LangSplat and ConceptGraphs smoke baselines and documented them as setup/adaptor evidence only.
- Audited public dataset readiness and confirmed that official Replica and ScanNet evaluations are not yet present.
- Updated the paper source and rebuilt the article PDF around the correct claim: strong cost reduction with a measured quality trade-off.
- Created a publication venue fit assessment recommending TwinWorld first and WACV next.

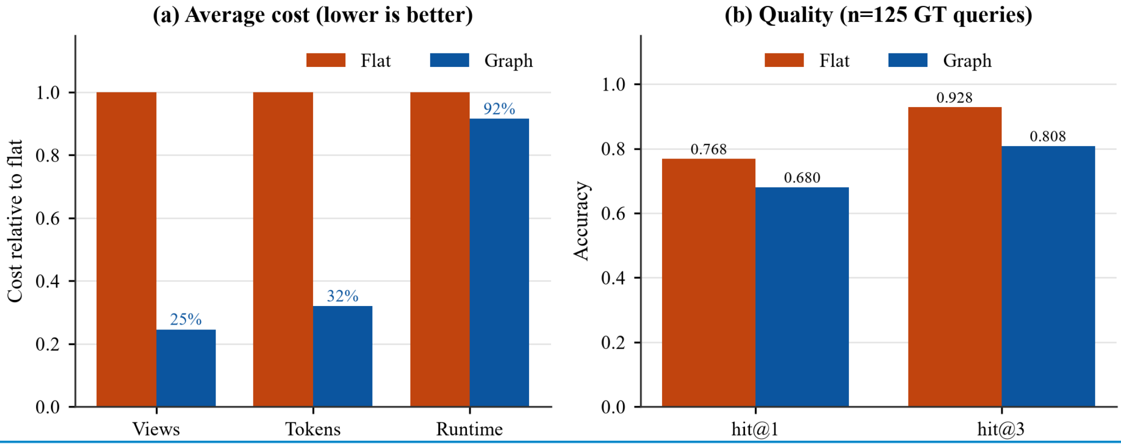
Current Evidence Snapshot

Artifact	Status	Evidence path
Five-scene benchmark	DONE	outputs/graph_vs_flat/five_scene_graph_vs_flat_v2/
Ablation study	DONE	outputs/graph_vs_flat/ablations_v1/
Scaling stress test	DONE	outputs/graph_vs_flat/scaling_stress_v1/
LangSplat baseline	SMOKE ONLY	outputs/baselines/langsplat_smoke_v1/
ConceptGraphs baseline	SMOKE ONLY	outputs/baselines/conceptgraphs_smoke_v1/
Public Replica	NOT DONE	docs/datasets/public_dataset_readiness.md
ScanNet	NOT DONE	docs/datasets/public_dataset_readiness.md
Article PDF	UPDATED	papers/beyond-proximity/main.pdf

2. Main Quantitative Result

The central result is a same-input comparison: graph and flat methods use the same scenes, semantic items, query strings, and verified view labels. This isolates the effect of hierarchical pruning.

Metric	Flat	Graph	Interpretation
Views checked	19.40	4.77	75.5% fewer views
Estimated input tokens	3168.2	1014.3	68.2% fewer tokens
Context chars	12701.4	4084.8	Less query-time context
Infra elapsed ms	6.61	6.05	Small local runtime difference
hit@1	0.768	0.680	Graph lower than flat
hit@3	0.928	0.808	Graph lower than flat

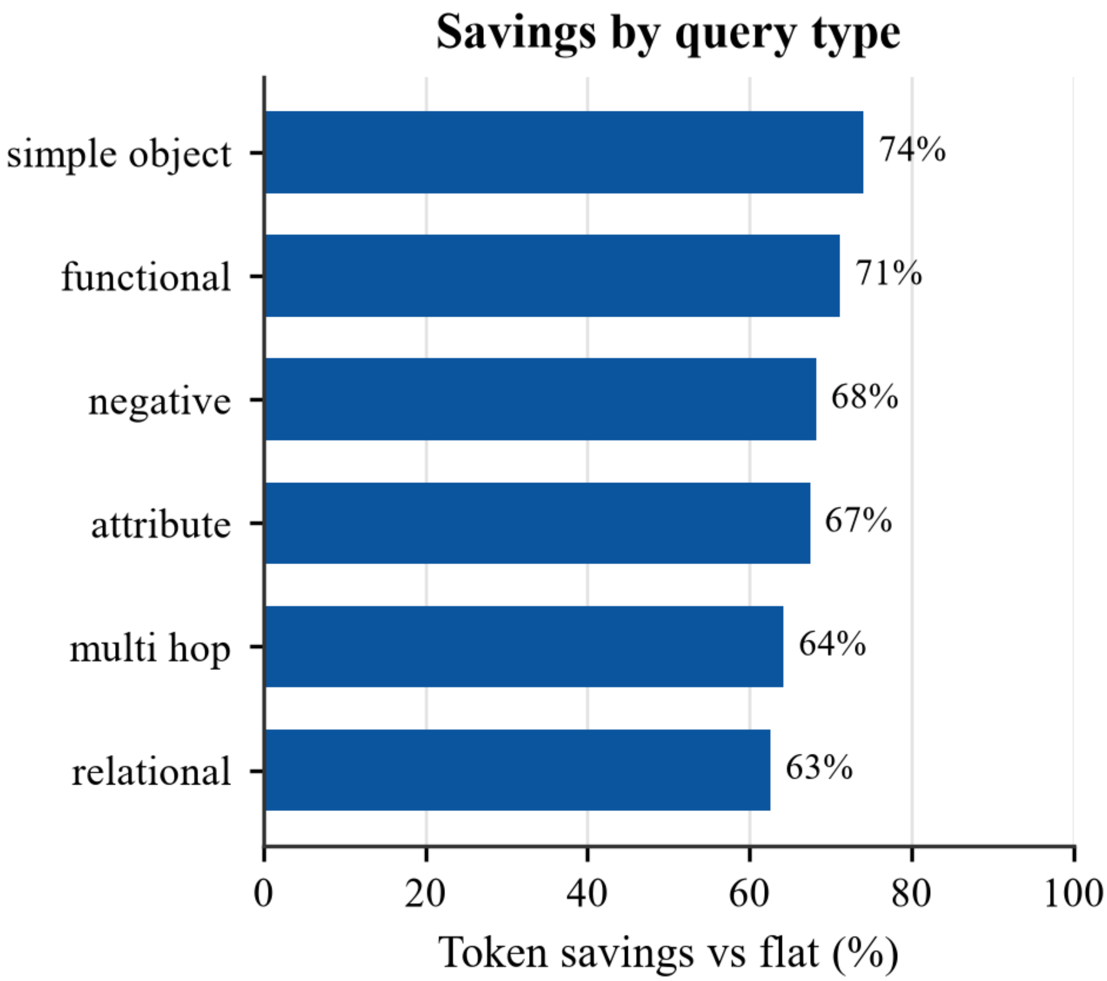


Correct interpretation: SemanticSplat currently proves cost control, not superiority. The graph prunes aggressively and saves context, but under the lexical stub ranker it also loses some correct candidates.

3. Ablations And Scaling

The ablation study gives the next algorithmic direction. Tight graph traversal saves the most cost but hurts hit@k; broad traversal recovers recall but loses token efficiency. The project needs calibrated pruning or confidence-based fallback.

Variant	Views	Token savings	hit@1	hit@3	Takeaway
flat_lexical	19.40	0.0%	0.768	0.928	Current exhaustive baseline
flat_affordance	19.40	0.0%	0.784	0.944	Best current quality reference
graph_tight	2.07	85.77%	0.600	0.720	High savings, too much pruning
graph_default	4.77	67.99%	0.680	0.808	Current paper setting
graph_broad	19.40	-100.33%	0.728	0.920	Recall improves but cost doubles
graph_affordance	4.99	66.47%	0.680	0.816	Small hit@3 gain



Scaling Stress Test

Multiplier	Flat views	Graph views	View savings	Token savings
1x	20.0	4.32	78.42%	71.20%
2x	40.0	7.43	81.42%	75.38%
5x	100.0	16.78	83.22%	77.89%
10x	200.0	32.37	83.82%	78.73%

Guardrail: the scaling study duplicates existing semantic views in memory. It is useful query-time stress evidence, not a substitute for Replica or ScanNet.

4. What Is Still Limited

Limitation	How serious?	Do we need to fix it?
Manual/reference labels, not independent GT	Medium for workshop, high for main conference	Yes before WACV/3DV/AAAI if making accuracy claims.
Graph hit@k lower than flat	High for algorithmic claims	Yes. Add calibrated pruning, fallback expansion, or embedding node scoring.
LangSplat and ConceptGraphs are smoke only	Acceptable for workshop scope, weak for main conference	Yes before claiming baseline comparison.
No official Replica/ScanNet evaluation	Medium for TwinWorld, high for WACV/3DV/AAAI	Replica first; ScanNet second after data access and converter work.
No 3D IoU or metric localization	High for 3D localization/navigation claims	Only needed if the paper claims object boxes, metric grounding, or navigation.
Article is not anonymized or venue-formatted	Blocking for submission	Yes. Prepare review PDF in the selected template.

Publication Fit

Venue	Current fit	After realistic improvements	Decision
TwinWorld @ ECCV 2026	78/100	88/100	Best immediate target
WACV 2027	66/100	84/100	Best main-conference path
3DV 2027	61/100	80/100	Good after stronger 3D grounding
NeurIPS 2026 workshops	55/100	75/100	Opportunistic backup
ICRA 2027	43/100	70/100	Needs embodied-agent evaluation
AAAI 2027 main	38/100	68/100	Too risky now

5. Next Steps

Immediate TwinWorld Sprint: 2026-07-06 to 2026-07-31

Priority	Task	Owner type	Output
P0	Choose venue strategy: short 4-page vs full 8-10-page TwinWorld paper.	Team decision	Submission plan
P0	Convert paper to ECCV LNCS style and anonymize.	Paper lead	Review PDF
P0	Add system overview and query-flow figures.	Figures/UX lead	Two paper figures
P0	Add real scene screenshots or graph overlays.	Data/visual lead	At least one qualitative figure
P0	Insert ablation table and explain cost/quality trade-off.	Evaluation lead	Updated experiments section
P1	Freeze commit hash, run IDs, and reproduction commands.	Engineering lead	Supplementary README
P1	Prepare demo/project page or short video if time allows.	Demo lead	Submission supplement

Main-Conference Research Sprint: August 2026

- Add a flat embedding baseline and compare it to flat lexical, graph default, and graph affordance.
- Implement calibrated graph pruning or low-confidence fallback expansion to recover hit@k while keeping cost low.
- Add official Replica evaluation if data can be legally obtained and documented.
- Turn ConceptGraphs and/or LangSplat from smoke evidence into fair same-scene baseline evidence where feasible.
- Add current five-scene failure analysis focused on missed graph candidates, wrong zones, and missing affordances.
- Prepare an anonymized supplementary package with code, query set, schemas, and reproduction commands.

Decisions Needed From The Team

Decision	Recommended answer
Where to submit first?	TwinWorld @ ECCV 2026 if a workshop publication is acceptable.
Should the TwinWorld paper be short or full?	Short if preserving future main-conference novelty matters; full if TwinWorld is the main target.
Do we need Replica/ScanNet now?	Replica is the first public-data priority; ScanNet comes after data access and converter completion.
Can we claim superiority over LangSplat/ConceptGraphs?	No. Current evidence supports smoke execution only.
Can we say graph preserves quality?	No. Current evidence shows lower hit@k; say cost/quality trade-off.

6. Final Recommendation

The project is in a good place for a careful workshop paper. It has real scenes, a reproducible benchmark, ablations, scaling stress evidence, baseline smoke tests, and an honest paper draft. The strongest current claim is not that the graph is more accurate; it is that hierarchy substantially reduces context/search cost and exposes a clear recall-cost frontier.

For TwinWorld, polish the paper and show the system clearly. For WACV/3DV, improve the science: add embedding retrieval, calibrated pruning, public dataset evaluation, and fair external baselines. That is the path from a solid prototype paper to a serious main-conference submission.

Referenced Project Artifacts

- papers/beyond-proximity/main.tex
- papers/beyond-proximity/main.pdf
- PUBLICATION_VENUE_FIT_ASSESSMENT.md
- docs/reports/final/claim_audit.md
- docs/reports/final/research_upgrade_status.md
- docs/datasets/public_dataset_readiness.md
- outputs/graph_vs_flat/five_scene_graph_vs_flat_v2/metrics_summary.json
- outputs/graph_vs_flat/ablations_v1/ablation_summary.md
- outputs/graph_vs_flat/scaling_stress_v1/scaling_summary.md